



Version Control & Deployment

Git · GitHub · Branching · GitHub Pages

A Complete Instructor-Led Module

Git Basics

GitHub Repos

Branching

GitHub Pages



ICE BREAKER

Let's get to know each other!



Raise your hand if you have ever accidentally deleted a file and panicked!



What version control tool (if any) have you heard of before today?



Have you ever visited a website hosted on GitHub Pages?



Turn to your neighbor — share ONE thing you hope to learn today!

Learning Objectives

By the end of this lesson, you will be able to:

01

Explain what version control is and why it matters in software development

02

Use core Git commands to track changes, commit code, and manage history

03

Create and manage GitHub repositories for personal and team projects

04

Apply branching strategies to collaborate safely without breaking production code

05

Deploy a live website for free using GitHub Pages



What is Version Control?

Think of it as a 'time machine' for your code.

Version Control System (VCS): A tool that records changes to files over time so you can recall specific versions later, compare changes, and collaborate with others without overwriting each other's work.



Track History

Every change is logged with who made it, when, and why



Revert Mistakes

Roll back to any previous working state instantly



Collaborate

Multiple developers work simultaneously without conflict



Git Basics & Workflow

git init

Initialize a new local repository

git status

Check state of working directory

git add .

Stage all changes for commit

git log

View commit history

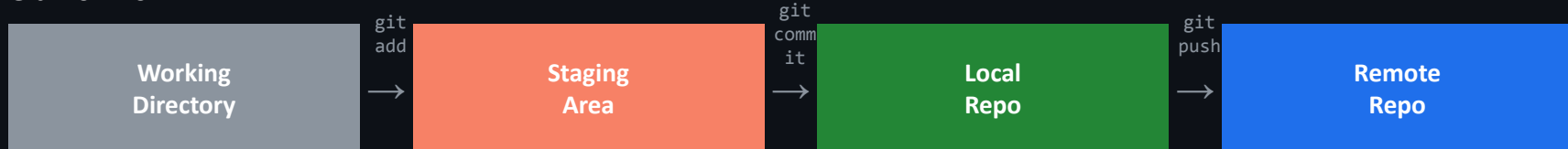
git commit -m "msg"

Save staged changes with a message

git clone <url>

Copy a remote repo locally

Git Workflow:





GitHub Repositories

GitHub is a cloud-based hosting service for Git repositories. It adds a web interface, collaboration tools, issue tracking, and CI/CD pipelines on top of Git.



Public:

Anyone can view & fork. Great for open source.



Private:

Only invited collaborators can access.

Key Repository Concepts



README.md

Project description, setup instructions, usage docs



.gitignore

Files/folders Git should NOT track (node_modules, secrets)



LICENSE

Legal terms for how others can use your code



Fork

Your personal copy of someone else's repository



Pull Request

Propose changes to be merged into the main branch



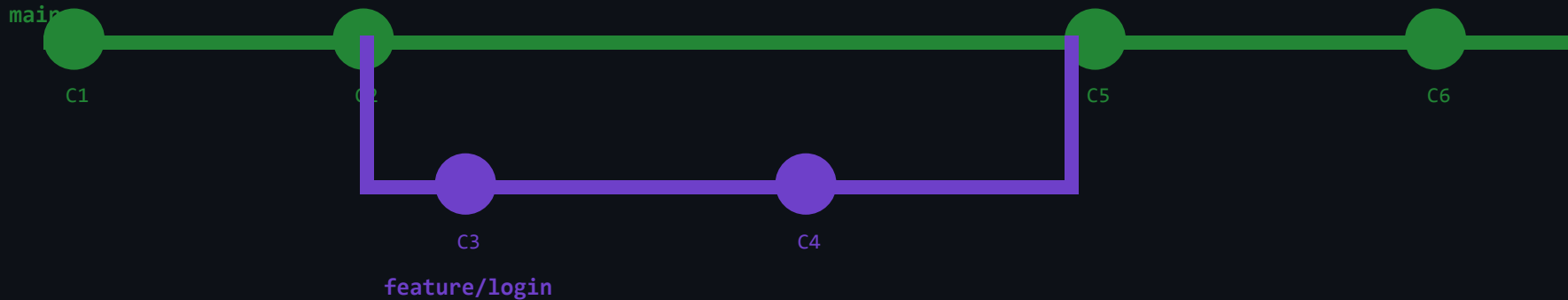
Stars & Watch

Bookmark repos and get notifications on updates



Branching & Collaboration Basics

Visual: Feature Branch Workflow



```
git branch <name>
```

Create a new branch

```
git checkout <name>
```

Switch to a branch

```
git checkout -b <name>
```

Create & switch in one step

```
git merge <name>
```

Merge branch into current

```
git pull origin main
```

Sync with remote main



Deploying Websites using GitHub Pages

GitHub Pages = Free, instant web hosting straight from your repository!

1

Push your HTML/CSS/JS to GitHub

Your repo must contain an index.html in the root or /docs folder

2

Go to Repository Settings

Navigate to Settings → Pages in your GitHub repository

3

Select Source Branch

Choose 'main' branch (or gh-pages) as the source

4

Save & Wait ~1 minute

GitHub builds and deploys your site automatically

5

Visit Your Live Site!

URL format: `https://username.github.io/repository-name`



Real-World Scenarios & Examples



Tech Company (e.g., Google, Meta)

Git: Hundreds of engineers commit code daily. Git ensures no one overwrites each other. Feature branches keep unstable code away from production.

GitHub/Pages: Internal GitHub Enterprise with code reviews, CI/CD pipelines, and automated testing on every pull request.



Student Portfolio

Git: A student tracks every version of their capstone project. When a bug appears, they use git log to find the exact commit that broke things.

GitHub/Pages: Repo shared with professors for grading. GitHub Pages hosts the live portfolio at their-name.github.io



Open Source Project (e.g., React, VS Code)

Git: Thousands of contributors worldwide. Each contributor forks the repo, makes changes on a branch, then submits a Pull Request.

GitHub/Pages: Maintainers review PRs, leave comments, request changes, and merge approved contributions.



Importance & Advantages



Complete Audit Trail

Every change is recorded: who made it, when, and why. Perfect for debugging and compliance.



Seamless Team Collaboration

Multiple developers work on the same codebase simultaneously without conflicts.



Easy Rollback

Made a mistake? Revert to any previous state in seconds with `git revert` or `git reset`.



Safe Experimentation

Branches let you experiment freely. Break things on a branch without affecting production.



Free Web Hosting

GitHub Pages makes deploying static sites completely free with a custom URL.



Industry Standard

Git + GitHub is used by 90%+ of professional development teams worldwide.



Disadvantages & Common Challenges



Steep Learning Curve

Problem: Git commands can be intimidating for beginners. Concepts like rebasing, stashing, and detached HEAD states confuse newcomers.



Fix: Start with just 5 commands: `init`, `add`, `commit`, `push`, `pull`. Learn advanced features gradually.



Merge Conflicts

Problem: When two developers edit the same file differently, Git can't auto-merge, requiring manual resolution.



Fix: Communicate with teammates, use short-lived branches, and `pull/sync` frequently to minimize drift.



Committing Sensitive Data

Problem: Accidentally pushing API keys, passwords, or private data to a public GitHub repo is a serious security risk.



Fix: Always use `.gitignore` for secrets. Use environment variables. Services like GitGuardian can alert you.



Large File Handling

Problem: Git is not optimized for large binary files (videos, PSD files). Repos can become slow and bloated.



Fix: Use Git LFS (Large File Storage) or store large assets in dedicated storage (S3, Cloudinary).



Key Takeaways

1

Git is your safety net

Commit often, commit early. Every commit is a checkpoint you can return to.

2

Branches = Freedom

Never work directly on main. Create a branch for every feature or fix.

3

GitHub ≠ Git

Git is the tool; GitHub is the platform. You can use Git without GitHub, but not vice versa.

4

Pull Requests are conversations

PRs are not just code merges — they're code reviews, discussions, and quality gates.

5

GitHub Pages = Instant Portfolio

Any static HTML/CSS/JS project can be live on the web in under 5 minutes — for free.

6

Version control is a career skill

Employers expect Git fluency. It appears in nearly every junior developer job description.

Suggested Videos to Watch

Watch these to deepen your understanding:



Git and GitHub for Beginners – Crash Course

freeCodeCamp.org • 1hr 8min • [🔗 youtube.com/watch?v=RG0j5yH7evk](https://youtube.com/watch?v=RG0j5yH7evk)

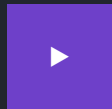
Git Basics & GitHub



Git Tutorial for Beginners: Learn Git in 1 Hour

Programming with Mosh • 1hr 9min • [🔗 youtube.com/watch?v=8JJ101D3knE](https://youtube.com/watch?v=8JJ101D3knE)

Git Workflow



GitHub Branches and Merging

The Coding Train • 32min • [🔗 youtube.com/watch?v=oPpnCh7lnLY](https://youtube.com/watch?v=oPpnCh7lnLY)

Branching & Collaboration



Deploy a Website with GitHub Pages

Traversy Media • 12min • [🔗 youtube.com/watch?v=SKXkC4SqRk](https://youtube.com/watch?v=SKXkC4SqRk)

GitHub Pages



References

Official Docs

- Git SCM Documentation — git-scm.com/doc
- GitHub Docs — docs.github.com
- GitHub Pages Docs — docs.github.com/pages

Books

- Pro Git (Free) — Chacon & Straub — git-scm.com/book
- Version Control with Git — Loeliger & McCullough (O'Reilly)

Tutorials & Courses

- Atlassian Git Tutorial — atlassian.com/git/tutorials
- The Odin Project: Git Basics — theodinproject.com
- freeCodeCamp: Version Control with Git — freecodecamp.org

Tools

- GitHub Desktop (GUI) — desktop.github.com
- GitKraken (GUI) — gitkraken.com
- VS Code Git Integration — code.visualstudio.com



You're Ready to Code with Confidence!

- ✓ Git tracks every change — your code is always safe
- ✓ GitHub connects you to the global developer community
- ✓ Branches let you experiment without fear
- ✓ GitHub Pages puts your work in front of the world — for free

"Every expert was once a beginner. Start committing today."